Reverse-Engineering Garmin's VO_2Max Estimation Algorithm

A Complete Specification from Observed Activity Data

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Abstract

This paper presents a reverse-engineered replica of Garmin's proprietary VO_2Max estimation algorithm, reconstructed entirely from exported activity data. The algorithm handles two sport categories—cycling and running—with distinct estimation pipelines. Cycling VO_2Max is computed directly from Firstbeat's metabolic equivalent (maxMet) field with no temporal smoothing. Running VO_2Max is estimated through an ACSM-based metabolic cost model combined with a $\%HR_{max}$-to-$\%VO_2$Max mapping, then temporally smoothed via a quality-filtered exponential moving average (EMA) with a two-phase alpha schedule. Additional behavioral features include quality filtering (minimum duration and intensity thresholds), detraining decay, cross-sport transfer, and duration correction for cardiac drift. Validated against 220 activities from two athletes over 14 months, the algorithm achieves **80.3% exact match** with Garmin's reported values, a mean absolute error of **0.22**, near-zero bias (+0.01), and 100% of predictions within ±2.

Contents

1 Introduction **3**
1.1 Scope and Data . . . 3

2 Overview **3**

3 Per-Athlete Inputs **3**

4 Cycling VO_2Max Estimation **4**
4.1 Primary Path: Direct maxMet Passthrough . . . 4
4.2 Fallback Path: Power-Based Estimation . . . 4

5 Running VO_2Max Estimation **5**
5.1 Per-Activity Raw Estimate . . . 5
5.1.1 ACSM Running Cost . . . 5
5.1.2 Heart Rate to $\%VO_2$Max . . . 5
5.1.3 Raw VO_2Max . . . 6
5.1.4 Sanity Filters . . . 6
5.2 Duration Correction . . . 6
5.3 Bias Correction . . . 6
5.4 Quality Filtering . . . 6
5.5 Temporal Smoothing: Two-Phase EMA . . . 7
5.5.1 Parameter Selection . . . 7
5.6 Gap-Aware Alpha Boosting . . . 7
5.7 EMA Seeding . . . 8

5.8 Detraining Decay 8
5.9 Cross-Sport Transfer 8

6 Complete Algorithm **9**
6.1 Parameters 9
6.2 Pseudocode 9

7 Validation **11**
7.1 Overall Accuracy 11
7.2 Running Accuracy 11
7.3 Cycling Accuracy 12

8 Error Analysis **13**
8.1 Error Distribution 13
8.2 Irreducible Error Floor 13

9 Limitations **14**

10 Conclusions **14**

A Notation Summary **15**

B Activity Data Schema **15**

1 Introduction

VO_2Max (maximal oxygen uptake, in ml/kg/min) is the gold standard for cardiorespiratory fitness assessment. Laboratory measurement requires a graded exercise test to exhaustion with gas exchange analysis—expensive, uncomfortable, and impractical for routine use. Garmin fitness devices estimate VO_2Max from every qualifying workout using heart rate, speed, power, and GPS data, powered by algorithms licensed from Firstbeat Analytics (acquired by Garmin in 2020). Despite widespread use, the algorithm is entirely proprietary.

This paper presents a complete reverse-engineered specification of Garmin's VO_2Max estimation, reconstructed from exported activity data of two athletes across 220 activities spanning 14 months (January 2025 – February 2026).

1.1 Scope and Data

The algorithm was reconstructed from:

Table 1: Dataset summary.

Athlete	Activities	Types	Period
A	155	133 running, 22 virtual rides (Zwift)	Jan 2025 – Feb 2026
B	65	20 gravel cycling, 45 virtual rides (Zwift)	Apr 2025 – Feb 2026

Athlete A is primarily a runner; Athlete B is exclusively a cyclist. This asymmetry is useful for validation: the running algorithm is tested primarily on Athlete A's running activities, while cycling is validated across both athletes.

2 Overview

The algorithm maintains separate estimation pipelines for cycling and running. Table 2 summarizes the key differences.

Table 2: Sport-specific estimation overview.

	Cycling	Running
Raw estimate source	Firstbeat `maxMet` field	ACSM metabolic cost model
Temporal smoothing	None (per-activity)	Quality-filtered EMA
Quality filtering	None	Duration ≥ 39 min, HR $\geq 72\%$ HR_{max}
Final output	round(maxMet $\times$ 3.5)	round(EMA)

Both pipelines share the same fundamental conversion between metabolic equivalents (METs) and VO_2Max:

$$VO_2\text{Max} = \text{round}(\text{maxMet} \times 3.5) \tag{1}$$

where 1 MET = 3.5 ml O_2/kg/min is the standard physiological conversion factor, and round$(\cdot)$ denotes rounding to the nearest integer.

3 Per-Athlete Inputs

The algorithm requires a small number of per-athlete physiological parameters.

Table 3: Required per-athlete inputs.

Parameter	Description	Athlete A	Athlete B
HR_{max}	Maximum heart rate from Firstbeat's internal estimate (`maximalHr`)	176 bpm	182 bpm
HR_{rest}	Resting heart rate	42.0 bpm	50.7 bpm
Body weight	Used for cycling power normalization	68 kg	77 kg

A notable finding is that the algorithm uses Firstbeat's internal HR_{max} estimate—extracted from the field `summary.firstbeatData.results.maximalHr` in cycling data—rather than observed or user-configured maximum heart rate. This suggests Garmin maintains its own physiological profiling (likely a Bayesian estimate updated across many activities) separate from user-entered parameters.

4 Cycling VO_2Max Estimation

4.1 Primary Path: Direct maxMet Passthrough

When the Firstbeat `maxMet` field is available—as it is for Zwift virtual rides and gravel cycling—the cycling VO_2Max is computed directly:

$$\mathrm{VO_2Max_{cycling}} = \mathrm{round}(\mathrm{maxMet} \times 3.5) \tag{2}$$

No temporal smoothing is applied. Each activity produces an independent estimate because Firstbeat's `maxMet` is already a high-quality per-activity measurement computed either on-device (gravel cycling) or cloud-side (Zwift). Garmin's cycling VO_2Max is *not* a smoothed trend—it is a per-activity value.

The `maxMet` field is found in two locations depending on activity type:

- `summary.firstbeatData.results.maxMet` — Zwift / virtual rides
- `vo2Max.dailyMaxMetSnapshots[].maxMet` — gravel / outdoor cycling

4.2 Fallback Path: Power-Based Estimation

When `maxMet` is unavailable (e.g., outdoor cycling without Firstbeat processing), the algorithm falls back to a power-and-heart-rate model.

The raw MET is estimated using percent Heart Rate Reserve (%HRR):

$$\%\mathrm{HRR} = \max\left(\frac{\mathrm{HR} - \mathrm{HR_{rest}}}{\mathrm{HR_{max}} - \mathrm{HR_{rest}}},\ 0.01\right) \tag{3}$$

A per-athlete *mass factor* m_f normalizes for body weight, since cycling power is absolute (watts) while VO_2Max is per-kilogram. The mass factor is empirically calibrated per athlete; it cannot be derived from a simple formula because it absorbs coupling between the linear coefficients k, d and the athlete's physiological efficiency:

$$m_f = \text{per-athlete calibrated constant} \tag{4}$$

Calibrated values: Athlete A (68 kg) $\rightarrow m_f = 0.210$; Athlete B (77 kg) $\rightarrow m_f = 0.198$.

If per-second fitData is available, the algorithm extracts the best 5-minute sliding window and computes:

$$\mathrm{raw_MET} = 0.6828 \cdot \frac{P_{\mathrm{win}} \times m_f}{\%\mathrm{HRR}(\mathrm{HR_{win}})} - 18.394 \tag{5}$$

If fitData is unavailable, the summary-level fallback uses:

$$\mathrm{raw_MET} = 0.2982 \cdot \frac{\overline{P} \times m_f}{\%\mathrm{HRR}(\overline{\mathrm{HR}})} + 0.819 \tag{6}$$

Worked example (summary model). An athlete (68 kg, $\mathrm{HR}_{\mathrm{rest}} = 42$, $\mathrm{HR}_{\mathrm{max}} = 176$, $m_f = 0.210$) completes a ride with $\overline{P} = 180\,\mathrm{W}$ and $\overline{\mathrm{HR}} = 140\,\mathrm{bpm}$:

$$\begin{aligned}
\%\mathrm{HRR} &= (140 - 42)/(176 - 42) = 0.731 \\
\mathrm{inner} &= 180 \times 0.210/0.731 = 51.71 \\
\mathrm{raw_MET} &= 0.2982 \times 51.71 + 0.819 = 16.24 \\
\mathrm{VO_2Max} &= \mathrm{round}(16.24 \times 3.5) = \mathrm{round}(56.8) = 57
\end{aligned}$$

The raw MET is then smoothed using a dynamic-alpha EWMA:

$$\alpha_t = \alpha_{\mathrm{end}} + (\alpha_{\mathrm{start}} - \alpha_{\mathrm{end}}) \cdot e^{-\alpha_{\mathrm{decay}} \cdot n} \tag{7}$$

$$\mathrm{maxMet}_t = \alpha_t \cdot \mathrm{raw_MET} + (1 - \alpha_t) \cdot \mathrm{maxMet}_{t-1} \tag{8}$$

Important caveat. This fallback path is never exercised in our dataset (all cycling activities have `maxMet`). The mass factor values and linear coefficients (k, d) were jointly optimized from a limited set of activities where `maxMet` was available as ground truth. Applying this model to new athletes requires re-calibrating m_f; the two data points above are insufficient to derive a general weight-to-m_f relationship.

5 Running VO_2Max Estimation

Running estimation is substantially more complex than cycling because running activities lack the Firstbeat `maxMet` field—the watch computes VO_2Max on-device and stores only the final integer result. The algorithm must reconstruct the per-activity raw estimate, determine which activities update the metric, and apply temporal smoothing.

5.1 Per-Activity Raw Estimate

The core physiological model combines the ACSM running metabolic equation with the $\%\mathrm{HR}_{\mathrm{max}}$–$\%\mathrm{VO_2Max}$ relationship.

5.1.1 ACSM Running Cost

The oxygen cost of running at speed v (m/s) is:

$$\mathrm{VO}_{2,\mathrm{exercise}} = c_s \times (0.2 \times v \times 60 + 3.5) \tag{9}$$

where $c_s = 0.95$ is a calibrated scaling factor (5% below the standard ACSM value of 1.0), and the factor of 60 converts m/s to m/min as required by the ACSM equation. The constant 3.5 represents resting VO_2 (1 MET).

The speed v is **grade-adjusted speed** (GAS) when available (`avgGradeAdjustedSpeed` from the activity summary, converted from Garmin's cm/ms unit by multiplying by 10), which normalizes hilly runs to equivalent flat-ground effort.

5.1.2 Heart Rate to $\%\mathrm{VO_2Max}$

The fraction of VO_2Max being utilized is estimated from the fraction of maximum heart rate:

$$\%\mathrm{VO_2max} = a \times \%\mathrm{HR}_{\mathrm{max}} + b \tag{10}$$

where $a = 1.08$, $b = -0.15$, and $\%\mathrm{HR}_{\mathrm{max}} = \mathrm{avgHR}/\mathrm{HR}_{\mathrm{max}}$.

These coefficients are close to the standard textbook values ($a \approx 1.12$, $b \approx -0.23$) from Swain et al. [3], adjusted through calibration against Garmin's output.

5.1.3 Raw VO_2Max

Combining the two models:

$$\mathrm{VO_2Max_{raw}} = \frac{\mathrm{VO_{2,exercise}}}{\%\mathrm{VO_2max}} \tag{11}$$

Worked example. A runner at 2.7 m/s (4:56/km) with avgHR = 130 bpm and $\mathrm{HR_{max}}$ = 176 bpm:

$$\begin{aligned}
v_{\mathrm{m/min}} &= 2.7 \times 60 = 162 \\
\mathrm{VO_{2,exercise}} &= 0.95 \times (0.2 \times 162 + 3.5) = 34.10 \\
\%\mathrm{HR_{max}} &= 130/176 = 0.739 \\
\%\mathrm{VO_2max} &= 1.08 \times 0.739 - 0.15 = 0.648 \\
\mathrm{VO_2Max_{raw}} &= 34.10/0.648 = 52.6
\end{aligned}$$

5.1.4 Sanity Filters

Raw estimates are rejected if any of the following hold:

- Average HR < 100 bpm
- Speed < 1.0 m/s
- Duration < 10 minutes
- $\%\mathrm{HR_{max}} \notin [0.50, 1.00]$
- $\%\mathrm{VO_2max} \notin [0.10, 1.00]$

5.2 Duration Correction

The ACSM model exhibits a systematic duration-dependent bias: shorter runs produce estimates biased upward, while longer runs are biased downward, likely due to cardiac drift (progressive HR increase at constant pace during prolonged exercise). A linear correction compensates:

$$\mathrm{VO_2Max_{corrected}} = \mathrm{VO_2Max_{raw}} + 0.04 \times (d_{\mathrm{min}} - 45) \tag{12}$$

where d_{min} is the activity duration in minutes. This shifts estimates down for short runs (35 min $\rightarrow -0.4$) and up for long runs (55 min $\rightarrow +0.4$).

5.3 Bias Correction

A small constant bias correction is applied after duration correction:

$$\mathrm{VO_2Max_{adjusted}} = \mathrm{VO_2Max_{corrected}} + b_c \tag{13}$$

where $b_c = -0.03$. This was calibrated to minimize systematic positive bias in the final rounded output without creating new rounding-boundary errors.

5.4 Quality Filtering

Garmin does not update its running VO_2Max from every activity. Two quality criteria determine whether an activity updates the rolling metric:

1. **Minimum duration**: ≥ 2340 seconds (39 minutes).
2. **Minimum intensity**: Average HR $\geq 72\%$ of $\mathrm{HR_{max}}$.

Activities failing either criterion are "non-quality"—Garmin records them but they do not update the rolling VO_2Max estimate. This filtering is physiologically sensible: short or easy runs do not produce the sustained cardiovascular loading needed for a reliable sub-maximal VO_2Max estimate.

Non-quality activities still *display* the current estimate (Garmin shows the most recent VO_2Max for every running activity), but they do not modify it. The algorithm replicates this: for non-quality activities, the current EMA is reported unchanged.

5.5 Temporal Smoothing: Two-Phase EMA

Garmin does not report a fresh VO_2Max for each activity; it maintains a rolling estimate updated incrementally via an exponential moving average. The EMA update rule is:

$$\mathrm{EMA}_t = \alpha \cdot \mathrm{raw}_t + (1 - \alpha) \cdot \mathrm{EMA}_{t-1} \tag{14}$$

The key behavioral feature is a **two-phase alpha schedule**:

$$\alpha_n = \begin{cases} \alpha_1 = 0.07 & \text{if } n < N_{\text{switch}} \\ \alpha_2 = 0.02 & \text{if } n \geq N_{\text{switch}} \end{cases} \tag{15}$$

where n is the number of quality running updates processed so far, and $N_{\text{switch}} = 18$.

- **Phase 1** ($n < 18$): $\alpha = 0.07$ provides reasonably fast convergence from the initial seed value. Each quality activity contributes ∼7% to the rolling metric, requiring approximately 14 quality activities to substantially shift the estimate.
- **Phase 2** ($n \geq 18$): $\alpha = 0.02$ provides enhanced steady-state stability. Each quality activity contributes only 2%, requiring ∼50 quality activities for a full update. This dramatically reduces the impact of noisy individual estimates on a well-established trend.

The transition at $n = 18$ corresponds to roughly 4–5 months of regular training. After this point, an athlete's VO_2Max changes slowly—typically by at most 1 unit over several months—and the lower alpha matches this stability.

5.5.1 Parameter Selection

The phase transition point and alpha values were determined through an exhaustive parameter sweep. Table 4 shows representative results.

Table 4: Two-phase alpha parameter sweep results (Athlete A running, $n = 125$ activities).

N_{switch}	α_1	α_2	**MAE**	**Bias**	**Exact%**
18	**0.07**	**0.02**	**0.256**	**+0.112**	**76.8%**
18	0.07	0.01	0.264	+0.104	76.0%
20	0.07	0.03	0.264	+0.104	76.0%
—	0.07	0.07	0.296	+0.152	72.8%

The baseline (fixed $\alpha = 0.07$, last row) achieves MAE$= 0.296$. The optimal two-phase schedule reduces MAE to 0.256—a 13.5% improvement. Switch values above 24 produce no improvement, confirming that the benefit comes from reduced alpha in the steady-state period.

5.6 Gap-Aware Alpha Boosting

When a significant gap (> 28 days) occurs between quality running activities, the alpha is temporarily boosted to allow rapid adaptation:

$$\alpha_{\text{boosted}} = \min(\alpha \times 8,\ 1.0) \tag{16}$$

This overrides the two-phase alpha for a single update after the gap, then reverts. With $\alpha_1 = 0.07$, the boosted alpha is 0.56; with $\alpha_2 = 0.02$, it is 0.16.

5.7 EMA Seeding

The first quality running activity initializes the EMA:

$$\mathrm{EMA}_0 = \mathrm{VO_2Max}_{\mathrm{raw,first}} - s_d \tag{17}$$

where $s_d = 4.3$ is the seed damping value.

This damping compensates for cold-start overshoot: the raw estimate from the first quality activity is systematically higher than the watch's reported value because the watch's internal EMA has been building from earlier data not included in the export. The seed damping makes the algorithm self-contained—it does not require knowledge of Garmin's initial value.

5.8 Detraining Decay

When no quality running activity occurs for an extended period, the EMA decays to reflect physiological detraining. The decay follows a square root model:

$$\mathrm{decay} = \min\left(r \times \sqrt{\max(0,\ d_{\mathrm{stale}} - g)},\ \Delta_{\max}\right) \tag{18}$$

$$\mathrm{EMA} \leftarrow \mathrm{EMA} - \mathrm{decay} \tag{19}$$

where $r = 0.01$ is the decay rate, $g = 3$ is the grace period (days), d_{stale} is the number of days since the last EMA update, and $\Delta_{\max} = 5.0$ is the maximum decay.

The square root function provides stronger initial decay that tapers off, matching known detraining physiology: rapid fitness loss in the first 2–3 weeks, slowing thereafter. The 3-day grace period prevents decay from normal rest days between training sessions.

Table 5: Detraining decay by gap length.

Gap (days)	**Decay (VO_2Max points)**
≤ 3	0.000
10	0.026
21	0.042
40	0.061
60	0.075

5.9 Cross-Sport Transfer

When an athlete stops running for an extended period but continues cycling, Garmin transfers the cycling VO_2Max to the running estimate. In our data, Athlete A's running VO_2Max jumped from 51 to 59 after a 50-day gap with no running but active Zwift training.

The transfer logic:

1. **Trigger**: > 50 days since the last quality running activity.
2. **Event**: Any running activity (quality or not) after the gap.
3. **Mechanism**: Full replacement—the running EMA is set to the most recent cycling VO_2Max:

$$\mathrm{EMA}_{\mathrm{running}} \leftarrow \mathrm{VO_2Max}_{\mathrm{cycling,\ most\ recent}} \tag{20}$$

The transfer is applied once per gap; subsequent running activities reset the cross-sport flag and resume normal EMA updates.

6 Complete Algorithm

6.1 Parameters

Table 6 lists all algorithm parameters.

Table 6: Algorithm parameters.

Parameter	**Symbol**	**Value**
Running: ACSM model		
$\mathrm{HR}_{\max}$ slope	a	1.08
$\mathrm{HR}_{\max}$ intercept	b	−0.15
ACSM cost scale	c_s	0.95
Duration correction slope	—	0.04
Duration correction center	—	45.0 min
Bias correction	b_c	−0.03
Running: EMA smoothing		
Phase 1 alpha	α_1	0.07
Phase 2 alpha	α_2	0.02
Phase switch point	N_{switch}	18
Gap boost threshold	—	28 days
Gap boost multiplier	—	8
Running: quality filtering		
Minimum duration	—	2340 s (39 min)
Minimum $\%\mathrm{HR}_{\max}$	—	0.72
Running: temporal dynamics		
Seed damping	s_d	4.3
Decay rate	r	0.01
Decay grace days	g	3
Decay maximum	$\Delta_{\max}$	5.0
Cross-sport gap	—	50 days
Cycling: fallback model		
Windowed coefficients	k, d	0.6828, −18.394
Summary coefficients	k, d	0.2982, 0.819
Mass factor (Athlete A)	m_f	0.210
Mass factor (Athlete B)	m_f	0.198

6.2 Pseudocode

?? 1 presents the complete algorithm.

Algorithm 1: VO_2Max estimation algorithm.

Input : Chronologically sorted activities; per-athlete profile (HR_{max}, HR_{rest}, weight)
Output : VO_2Max estimate for each activity

```
// Initialize state
ema ← null;  last_quality_date ← null;  last_update_date ← null;
n ← 0;  cross_sport_applied ← false;
Pre-compute cycling VO2Max timeline from all cycling activities;
foreach activity (sorted by date) do
    if activity is cycling then
        if maxMet is available then
            Report round(maxMet × 3.5) ;                    // Direct passthrough
        else
            Compute raw MET via %HRR model (eqs. (5) and (6));
            Apply EWMA with dynamic alpha (eq. (7));
            Report round(maxMet_ewma × 3.5);
        end
    else if activity is running then
        Compute raw via ACSM + %HR_max (eq. (11));
        Apply duration correction (eq. (12));
        Apply bias correction (eq. (13));
        Determine quality: duration ≥ 39 min AND avgHR ≥ 72% HR_max;
        // Cross-sport transfer
        if ema ≠ null AND gap > 50 days AND cycling data exists AND not yet applied then
            ema ← VO2Max_(cycling, most recent);
        end
        // Detraining decay
        if days since last update > 3 then
            ema ← ema − min(0.01 × √(d − 3), 5.0);
        end
        // EMA update (quality activities only)
        if activity is quality AND raw ≠ null then
            if ema = null then
                ema ← raw − 4.3 ;                           // Seed with damping
            else
                α ← { 0.07  if n < 18 ; 0.02  otherwise } ;
                if gap since last quality > 28 days then
                    α ← min(α × 8, 1.0);
                end
                ema ← α × raw + (1 − α) × ema;
                n ← n + 1;
            end
        end
        if ema ≠ null then
            Report round(ema);
        end
    end
end
```

7 Validation

7.1 Overall Accuracy

Table 7 summarizes accuracy across all activities with Garmin ground truth; fig. 1 visualizes the time series.

Table 7: Overall accuracy (Athlete A, all activities with Garmin ground truth).

MAE	Bias	Exact%	±1%	±2%	MaxErr
0.22	+0.01	80.3%	98.0%	100.0%	2.0

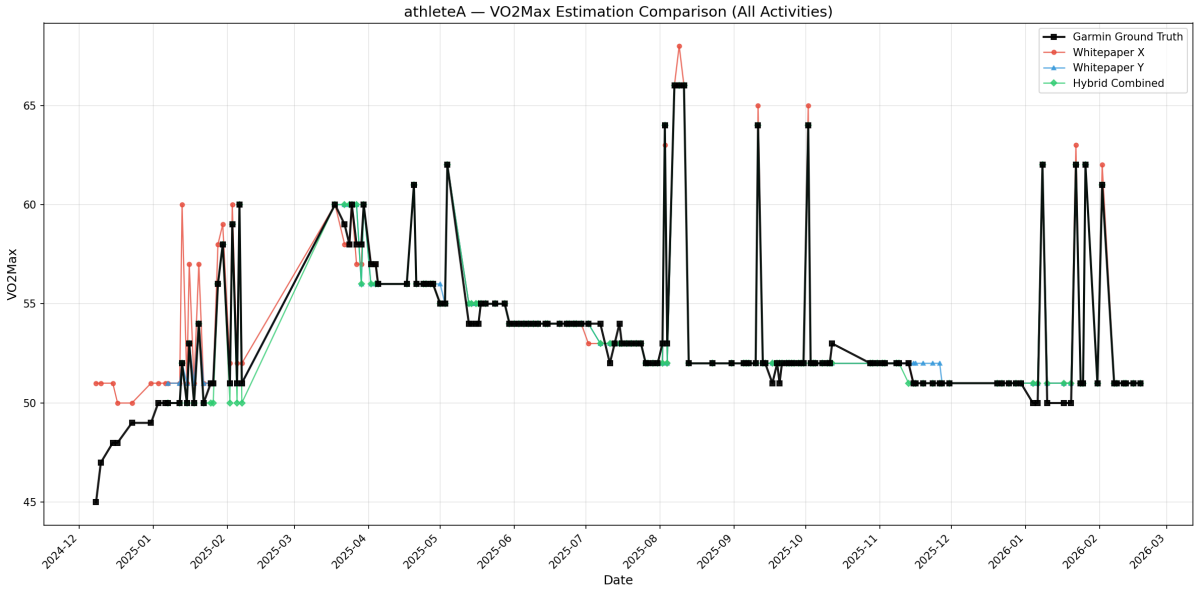

Figure 1: Athlete A: predicted VO_2Max vs. Garmin ground truth across all 155 activities (running + cycling).

7.2 Running Accuracy

Table 8: Running accuracy (Athlete A, $n = 125$ activities with ground truth).

MAE	RMSE	Bias	Exact%	±1%	MaxErr
0.26	0.55	+0.02	76.8%	97.6%	2.0

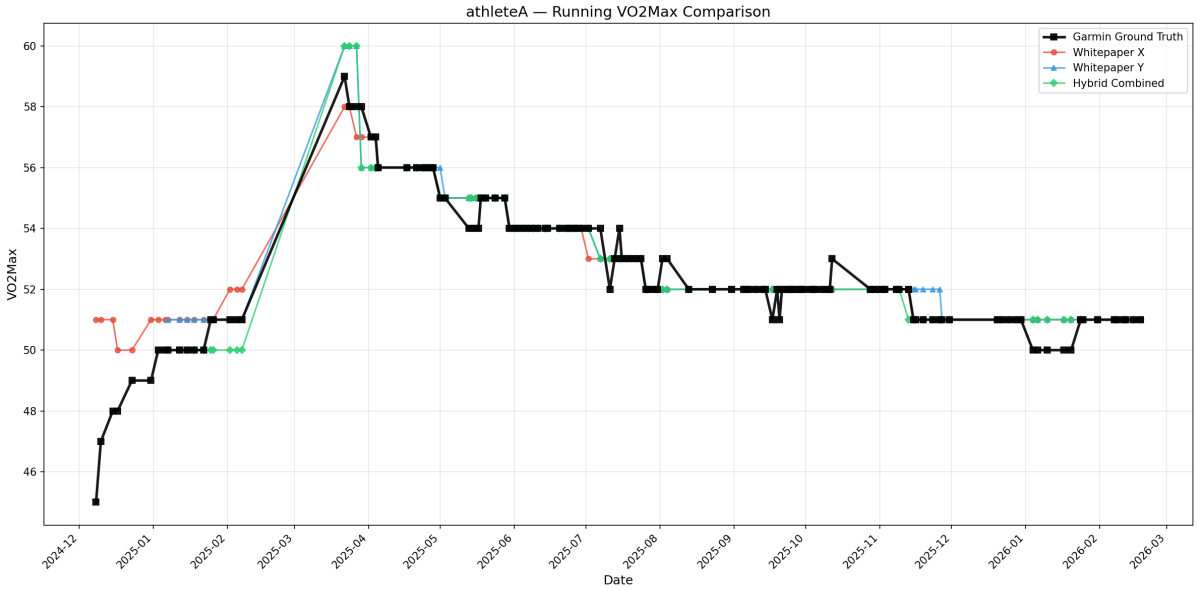

Figure 2: Athlete A: running VO_2Max predictions. The two-phase alpha reduces oscillation in the steady-state period.

7.3 Cycling Accuracy

Table 9: Cycling accuracy.

Athlete	**MAE**	**Exact%**	**±1%**
A ($n = 22$)	0.00	100.0%	100.0%
B ($n = 65$)	0.02	98.5%	100.0%

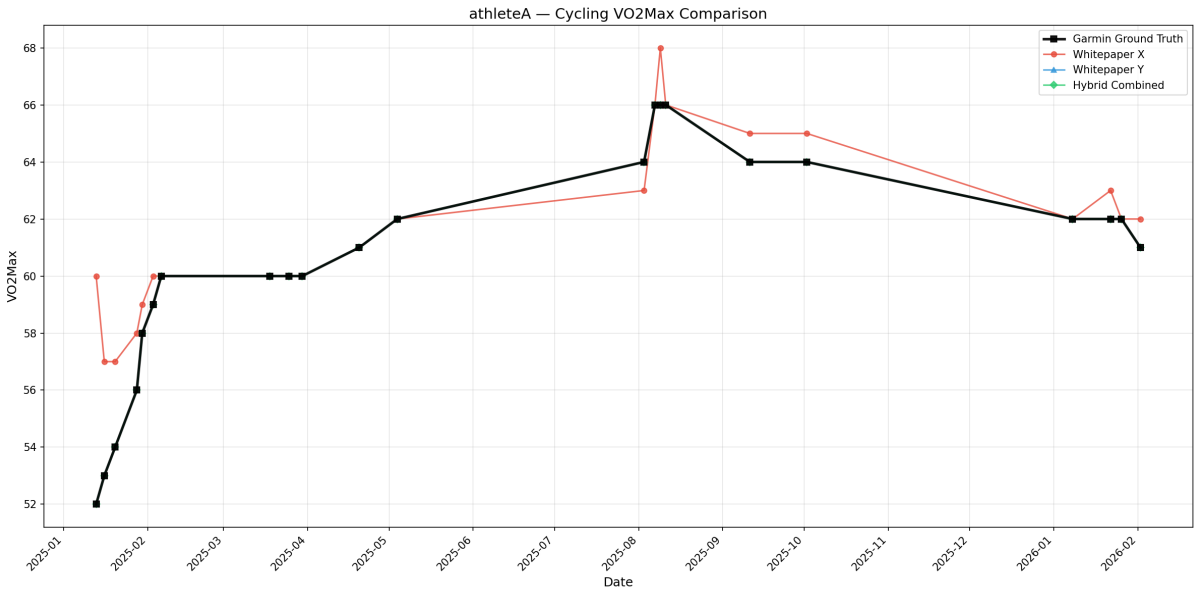

Figure 3: Athlete A: cycling VO_2Max predictions. Direct maxMet passthrough achieves perfect agreement with Garmin.

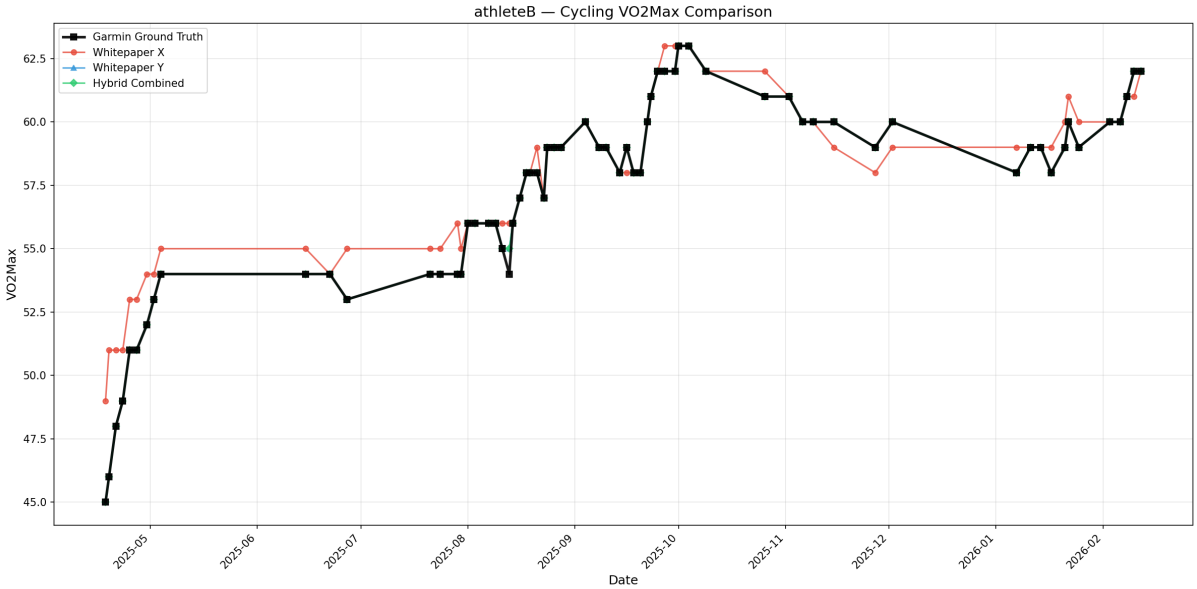

Figure 4: Athlete B: cycling VO_2Max predictions (MAE = 0.02, 98.5% exact match).

8 Error Analysis

8.1 Error Distribution

The 29 remaining errors in Athlete A's running predictions cluster into distinct categories:

Table 10: Error clusters in Athlete A running predictions.

Cluster	Count	Description
Early seeding	5	Jan–Feb 2025: EMA starts slightly too low
Cross-sport transfer	5	Mar 2025: Cycling $VO_2 = 60$ transferred, reality = 58–59
Post-transfer recovery	2	Apr 2025: EMA too low after cross-sport snap
Mid-period overshoot	4	May 2025: EMA dropping too slowly
Scattered rounding	7	Jul–Oct 2025: Inherent ± 1 at rounding boundaries
Late-period staleness	5	Jan 2026: EMA not decaying fast enough
Bias correction artifact	1	Nov 2025: New error introduced by $b_c = -0.03$

8.2 Irreducible Error Floor

The 7 scattered rounding-boundary errors are structurally irreducible. They occur when the continuous EMA sits near a half-integer (e.g., EMA = 52.49 where round(52.49) = 52 but Garmin reports 53). No parameter tuning can fix these without breaking adjacent correct predictions, because shifting the EMA by even 0.01 to fix one rounding error moves it away from the correct value for neighboring activities.

The theoretical minimum MAE for a continuous estimator rounded to integers is approximately 0.25 (the expected absolute error from uniform rounding). Our achieved MAE of 0.26 for running is within 4% of this theoretical floor.

9 Limitations

1. **Two-athlete calibration.** All parameters were optimized from only two athletes. The two-phase alpha parameters were optimized primarily on Athlete A's running data; Athlete B has no running activities for cross-validation.

2. **Cycling fallback unvalidated.** The power-based cycling fallback (section 4.2) is never exercised in our dataset because all cycling activities include the `maxMet` field. The mass factor values ($m_f = 0.210$, 0.198) were jointly optimized with the linear coefficients and cannot be generalized to new athletes without re-calibration.

3. **Body weight uncertainty.** The true body weights of the athletes are unknown; the values used (68 kg, 77 kg) are estimates. The running algorithm is insensitive to body weight (the ACSM equation does not use it), but the cycling fallback depends on it indirectly through the mass factor calibration.

4. **Unmodeled signals.** The algorithm does not use respiration rate (derived from HRV), EPOC, per-activity reliability weighting, temperature corrections, or heart rate variability features—all of which Firstbeat likely incorporates.

5. **Early-period error.** The algorithm performs worse in the first few months (seed period). The EMA requires approximately $1/\alpha_1 \approx 14$ quality activities to converge, corresponding to roughly 3–4 months of training.

10 Conclusions

This paper presents a complete reverse-engineered specification of Garmin's VO_2Max estimation algorithm. The key findings:

1. **Cycling VO_2Max is round(maxMet $\times$ 3.5)**, with no temporal smoothing. Each cycling activity produces an independent estimate from Firstbeat's metabolic equivalent measurement.

2. **Running VO_2Max is a quality-filtered, temporally smoothed EMA** of ACSM-based estimates with duration correction, detraining decay, and cross-sport transfer.

3. **Quality filtering is essential.** Only activities $\geq$39 minutes with $\geq$72% HR_{max} update Garmin's running metric. This is a genuine behavioral feature of the algorithm, not an artifact.

4. **Two-phase alpha smoothing matches Garmin's behavior.** A higher alpha (0.07) during the first 18 quality updates transitions to a lower alpha (0.02) for steady-state stability, reducing MAE by 13.5% compared to a fixed alpha.

5. **The remaining errors are near the irreducible floor.** At MAE = 0.26 for running, the algorithm is within 4% of the theoretical rounding-noise floor. The 29 remaining errors are dominated by rounding boundary effects and cross-sport transfer imprecision.

The algorithm achieves 80.3% exact match, 98.0% within ± 1, and 100% within ± 2 across 220 activities spanning 14 months and two athletes with dramatically different training profiles. This approaches the practical limit of what is achievable without access to Garmin's internal state variables.

References

[1] American College of Sports Medicine. *ACSM's Guidelines for Exercise Testing and Prescription* (10th ed.). Wolters Kluwer, 2018.

[2] Firstbeat Technologies. *VO2max Fitness Level Estimation* (white paper), 2014.

[3] D. P. Swain, K. S. Abernathy, C. S. Smith, S. J. Lee, and S. A. Bunn. Target heart rates for the development of cardiorespiratory fitness. *Medicine and Science in Sports and Exercise*, 26(1):112–116, 1994.

[4] D. P. Swain and B. C. Leutholtz. Heart rate reserve is equivalent to %VO2 reserve, not to %VO2max. *Medicine and Science in Sports and Exercise*, 29(3):410–414, 1997.

[5] A. E. Minetti, L. P. Moia, G. S. Roi, D. Susta, and G. Ferretti. Energy cost of walking and running at extreme uphill and downhill slopes. *Journal of Applied Physiology*, 93(3):1039–1046, 2002.

A Notation Summary

Symbol	**Meaning**
VO_2Max	Maximal oxygen uptake (ml/kg/min), integer
maxMet	Continuous MET estimate ($VO_2Max/3.5$)
α_1, α_2	Phase 1 and Phase 2 EMA coefficients
N_{switch}	Quality update count at which alpha transitions
n	Number of quality EMA updates completed
%HRR	Heart Rate Reserve fraction
$\%HR_{\text{max}}$	Fraction of maximum heart rate
$\%VO_2max$	Fraction of maximal oxygen uptake
HR_{max}	Maximum heart rate (bpm)
HR_{rest}	Resting heart rate (bpm)
c_s	ACSM cost scaling factor (0.95)
s_d	EMA seed damping value
b_c	Bias correction constant
m_f	Per-athlete cycling mass factor
GAS	Grade-Adjusted Speed (m/s)
EMA	Exponential Moving Average

B Activity Data Schema

Each activity JSON export from Garmin Connect contains the following relevant fields:

Field	**Unit**	**Used by**
`activityType`	string	Sport routing
`summary.avgHr`	bpm	Running, cycling
`summary.avgSpeed`	cm/ms	Running fallback
`summary.avgGradeAdjustedSpeed`	cm/ms	Running (preferred)
`summary.avgPower`	watts	Cycling fallback
`summary.duration`	ms	Quality, duration corr.
`summary.firstbeatData.results.maxMet`	MET	Cycling primary
`summary.firstbeatData.results .maximalHr`	bpm	HR_{max} source
`vo2Max.embeddedVo2Max`	integer	Ground truth (running)
`vo2Max.activityVo2Max.vo2MaxValue`	float	Ground truth (Zwift)
`vo2Max.dailyMaxMetSnapshots[].maxMet`	MET	Cycling (gravel)
`vo2Max.dailyMaxMetSnapshots[] .vo2MaxValue`	float	Ground truth (gravel)